Oficina teórico-prática de análises de dados - análise de invariância

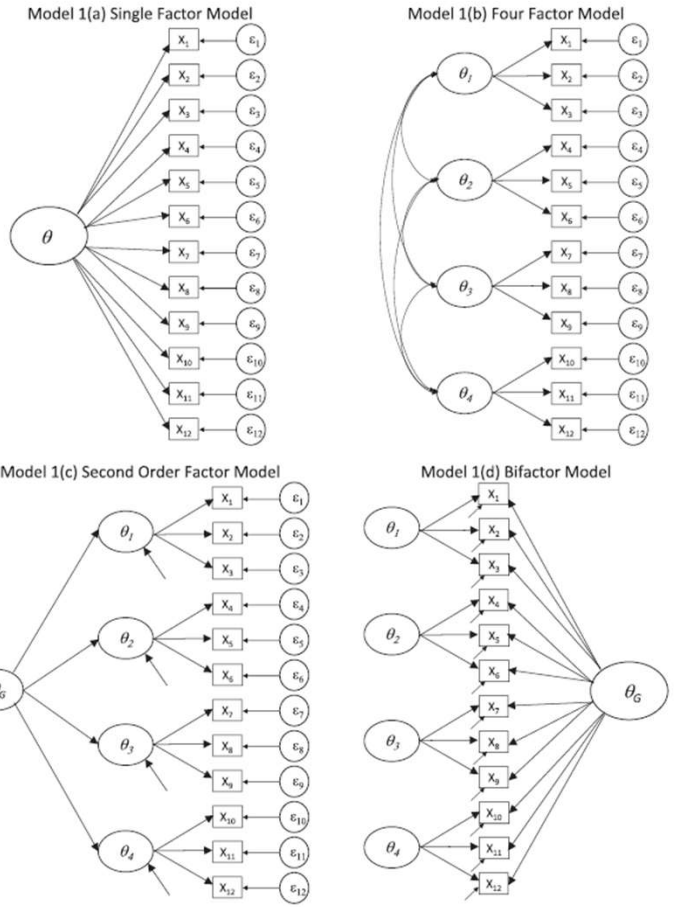
Pedro Martins

Roteiro

- Retomando conceitos dos modelos fatoriais
- Índices de ajuste
- O que é invariância?
- Diferentes níveis de invariância
- Qual a importância de estabelecer o nível de invariância antes da comparação de grupos?

O que é um modelo fatorial?

- A variável latente é a causa presumida da correlação observada entre os indicadores



Variável observável

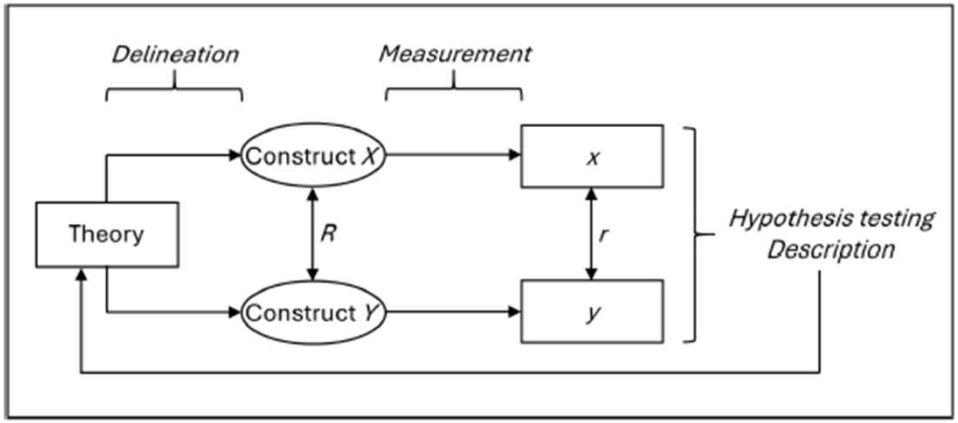
Podemos definir variáveis observáveis como aquelas que, de modo genérico, podemos ter razoável certeza de seu verdadeiro valor utilizando os sentidos. Ou seja, são variáveis que podem ser percebidas e a que temos acesso. Variáveis observáveis são representações do mundo empírico

Variável latente

Variáveis que, epistemologicamente, não temos acesso direto. São construções (construtos) teóricas que não podem ser observadas diretamente a partir dos sentidos. Diferença epistemológica

Construtos e medidas

Figure 2: Theories Specify Constructs and Their Relations



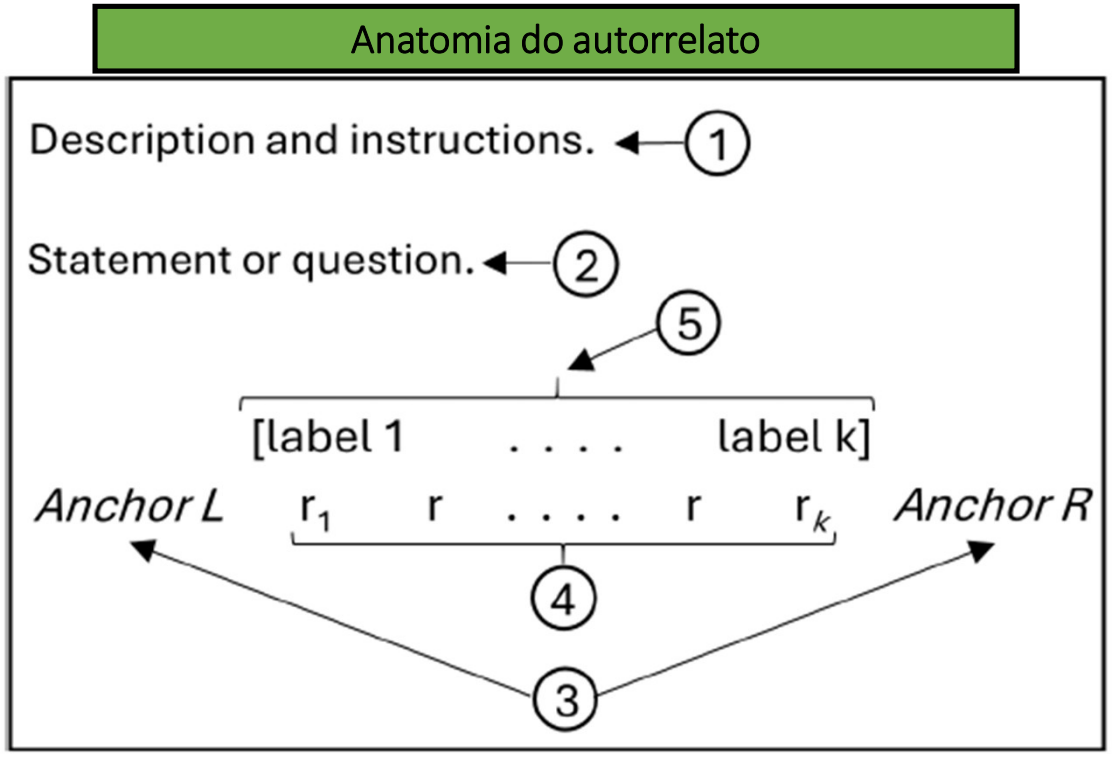
Note: Measurement is a primary means by which scores (x, y) are produced, enabling hypothesis testing (e.g., r) and description and informing theory.

O que são construtos?
Em termos amplos, um construto é alguma propriedade ou processo postulado, característico de pessoas, grupos, organizações, situações ou ambientes, que se assume estar refletido nos escores obtidos em medidas dos atributos dessa propriedade ou dos elementos desse processo. Os construtos são componentes importantes das teorias em psicologia (Shoemaker et al., 2004).

Construtos e medidas

- Formato mais comum na psicologia, pode assumir diferentes estratégias

1 – texto que orienta como responder aos itens
2 – Pergunta ou conteúdo dos itens
3 – Ancora determina o intervalo das avaliações (frequência, concordância, etc)
4 – Valores das âncoras, normalmente números entre 1 e 7
5 – Além dos rótulos das extremidades, todos os pontos podem ser nomeados



Como avaliar a qualidade dos modelos?

- Se estamos pensando apenas na qualidade da relação entre variáveis observáveis e latentes, usamos os chamados índices de ajuste (exatos e aproximados/relativos)

Teste	**Valor desejado**
Chi square (χ^2)	P valor maior que 0.05
Tucker-Lewis index (TLI)	Maior que 0.90 (ou maior que .95)
Comparative fit index (CFI)	Maior do que 0.90 (ou maior que .95)
Root mean square error of Approximation (RMSEA)	Menor que 0.08 (ou menor que 0.05)
Standardized root mean square residual (SRMR)	Menor que 0.08 (ou menor que 0.05)
Bayesian Information criterion (BIC)	Relativo, quanto menor melhor
Akaike Information criterion (AIC)	Relativo, quanto menor melhor

$$CFI = \frac{d_{null} - d_{specified}}{d_{null}} \text{ where } d = \chi^2 - df$$

$$RMSEA = \sqrt{\frac{(\chi_S^2 - df_S)/N}{df_S}} = \sqrt{\frac{(\chi_S^2/df_S) - 1}{N}}$$

But see

American Psychological Association

American Psychologist

© 2023 American Psychological Association

ISSN: 0003-066X

2023, Vol. 78, No. 9, 1061–1075
https://doi.org/10.1037/amp0001213

Dynamic Fit Index Cutoffs for Categorical Factor Analysis With Likert-Type, Ordinal, or Binary Responses

Daniel McNeish
Department of Psychology, Arizona State University

Scale validation is vital to psychological research because it ensures that scores from measurement scales represent the intended construct. Fit indices are commonly used to provide quantitative evidence that a proposed factor structure is plausible. However, there is a mismatch between guidelines for evaluating fit of the factor models and the data that most researchers have. Namely, fit guidelines are based on the simulations that assume item responses are collected on a continuous scale whereas most researchers collect discrete responses such as with a Likert-type scale. In this article, we show that common guidelines derived from assuming continuous responses (e.g., root-mean-square error of approximation < 0.06, comparative fit index > 0.95) do not generalize to factor models applied to discrete responses. Specifically, discrete responses provide less information than continuous responses, so less information about misfit is passed to fit indices. Traditional guidelines, therefore, end up being too lenient and lose their ability to identify that a model may have a poor fit. We provide one possible solution by extending the recently developed dynamic fit index framework to accommodate discrete responses common in psychology. We conduct a simulation study to provide evidence that the proposed method consistently distinguishes between well-fitting and poorly fitting models. Results showed that our proposed cutoffs maintained at least 90% sensitivity to misspecification across studied conditions, whereas traditional cutoffs were highly inconsistent and frequently exhibited sensitivity below 50%. The proposed method is included in the `dynamic` R package and as a web-based Shiny application to make it easily accessible to psychologists.

Public Significance Statement
Conclusions in many subareas of psychology are reliant on conclusions drawn from statistical models, which rely on assumptions. This article discusses how the assumptions made by statistical guidelines in the key area of scale validation differ from the data that most psychologists collect, at best making it difficult to evaluate the evidence supporting psychological scales and, at worst, increasing the probability that conclusions in many studies may be inaccurate. This directly feeds into replication issues observed in psychology because scale scores may not be accurately measuring psychological traits, attitudes, or abilities, which increases noise in the data and reduces the validity of scores. This article proposes alternative guidelines for one common type of statistical model by making assumptions more closely match the characteristics of the data to which scale validation models are applied.

Keywords: psychometrics, scale validation, CFA, categorical data analysis, SEM

Supplemental materials: https://doi.org/10.1037/amp0001213.supp

ANNUAL REVIEWS

Annual Review of Psychology

How Do Psychologists Determine Whether a Measurement Scale Is Good? A Quarter-Century of Scale Validation with Hu & Bentler (1999)

ANNUAL REVIEWS CONNECT
www.annualreviews.org
• Download figures
• Navigate cited references
• Keyword search
• Explore related articles
• Share via email or social media

Daniel McNeish
Department of Psychology, Arizona State University, Tempe, Arizona, USA; email: dmcneish@asu.edu

Annu. Rev. Psychol. 2026. 77:567–91

First published as a Review in Advance on August 19, 2025

The *Annual Review of Psychology* is online at psych.annualreviews.org

https://doi.org/10.1146/annurev-psych-121924-104021

Copyright © 2026 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

OPEN ACCESS

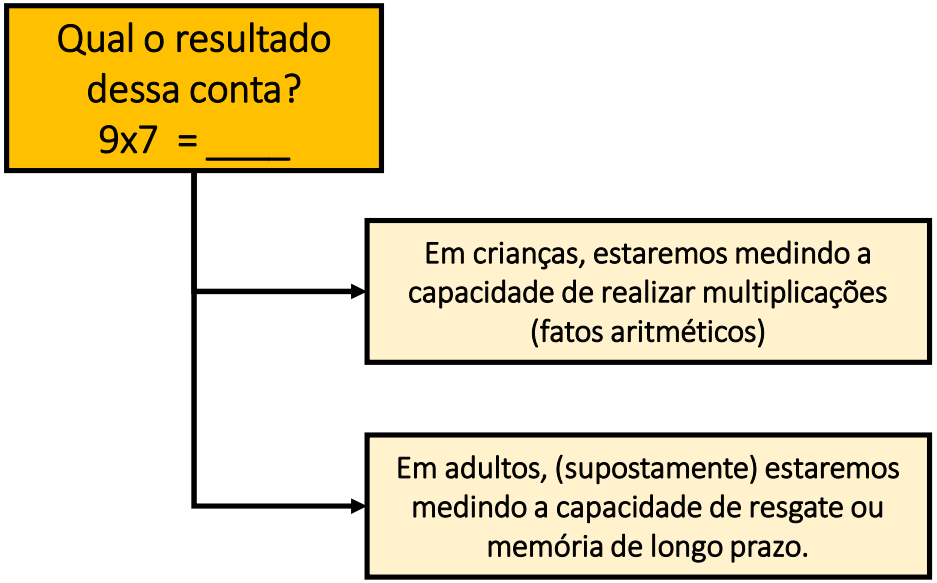
Keywords

measurement, model fit, structural equation modeling, psychometrics, validity, factor analysis

Abstract

Many psychologists rely on surveys, questionnaires, and measurement scales because psychological constructs like depression, motivation, or extraversion cannot be directly measured with physical instruments. Scale validation crucially provides evidence that scores from such scales capture their intended target. The prevailing scale validation approach involves comparing factor-analytic model fit indices to suggested benchmarks, and it is so engrained in psychological research that the article proposing the benchmarks is among the most cited works across any scientific discipline. However, methodological research finds that psychologists overgeneralize the benchmarks so that they no longer function as originally intended. This has widespread implications for psychologists and casts some doubt on conclusions regarding the validity of our measurement scales. This review covers the history and origin of scale validation benchmarks, how benchmarks rose to prominence and became overgeneralized, recently proposed alternatives to traditional

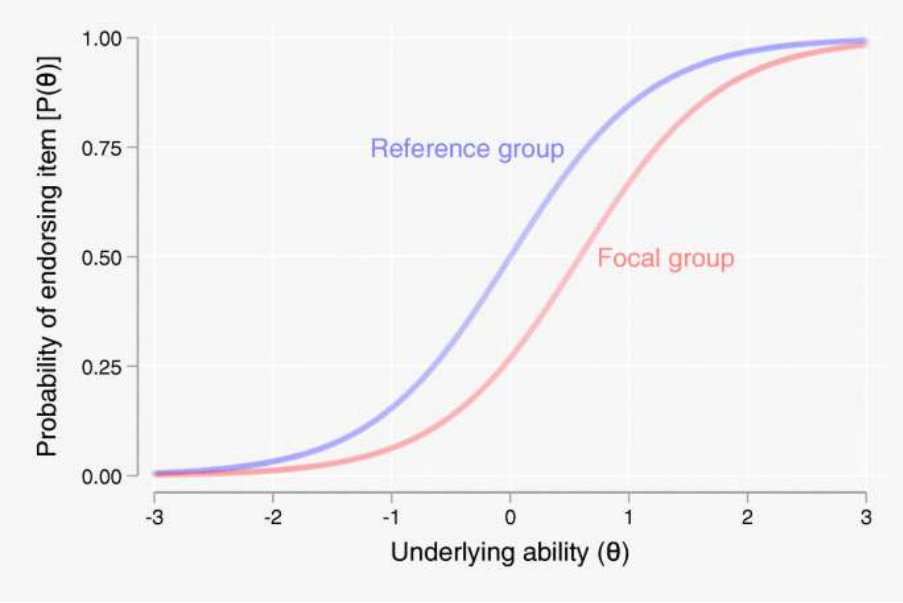
O que é invariância?



- Quando administramos um instrumento psicológico em diferentes grupos, ou em para as mesmas pessoas em momentos diferentes, queremos que o construto latente medido seja o mesmo!
- No exemplo, temos uma não invariância, o processo gerador da resposta não é o mesmo entre os grupos

O que é invariância?

- Invariância não é sinônimo de igualdade, mas de comparável
- Diferenças nas **pontuações totais** não são consideradas não invariantes do ponto de vista psicométrico [APENAS SE O PROCESSO GERADOR DO ESCORE NO ITEM FOR JUSTO – ou seja, sem DIF]



O que é invariância

Administração

Diferentes grupos recebem a mesma escala em condições similares

O construto existe?

Ajustar um modelo inicial que seja compatível para a amostra no geral

Constrangimento

Para buscar a equivalência entre os grupos, configuramos (1) estrutura, (2) cargas fatoriais, (3) interceptos (limiares), (4) erros, etc para serem iguais entre os grupos

Avaliação do ajuste

Avaliamos se fixar coisas como iguais piora o modelo. E quanto piora.

Reespecificação do modelo

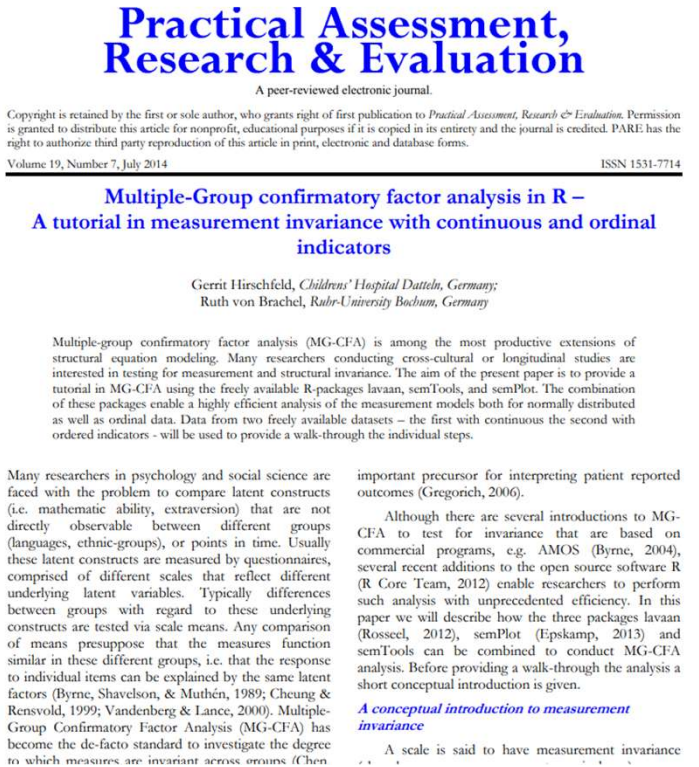
É possível verificar se apenas uma parte do modelo precisa ser “liberada” para que o ajuste fique razoável novamente.

Níveis de invariância

Nível da invariância	O que indica	Comparação possível	Como?
Configural	Os mesmos itens estão envolvidos no processo de medida dos mesmos contrutos	Nenhuma	Modelo inicial de grupos é aceitável
Métrica (weak, equal factors)	Cargas fatoriais iguais entre grupos. A relação entre o construto e os itens é o mesmo	Coeficientes de correlação e regressão não padronizados	O ajuste não piora na comparação com o modelo configural
Escalar (scalar, threshold, Strong)	Relações anteriores E que os interceptos ou limiares de resposta é o mesmo entre os grupos	Relações acima e médias	O ajuste não piora na comparação com o modelo métrico

Avaliando a piora

- Esses são valores convencionais!
- Diferenças aceitáveis em:
  - $\Delta \chi^2$ não significativa
  - Δ CFI $\leq 0{,}01$
  - Δ RMSEA $\leq 0{,}015$
- Para não confundir, não coloquei o sinal. Lembre que essa diferença é interpretada como negativa quando o indicador está piorando



Practical Assessment, Research & Evaluation

A peer-reviewed electronic journal.

Copyright is retained by the first or sole author, who grants right of first publication to *Practical Assessment, Research & Evaluation*. Permission is granted to distribute this article for nonprofit, educational purposes if it is copied in its entirety and the journal is credited. PARE has the right to authorize third party reproduction of this article in print, electronic and database forms.

Volume 19, Number 7, July 2014 ISSN 1531-7714

Multiple-Group confirmatory factor analysis in R – A tutorial in measurement invariance with continuous and ordinal indicators

Gerrit Hirschfeld, *Childrens' Hospital Datteln, Germany;*
Ruth von Brachel, *Ruhr-University Bochum, Germany*

Multiple-group confirmatory factor analysis (MG-CFA) is among the most productive extensions of structural equation modeling. Many researchers conducting cross-cultural or longitudinal studies are interested in testing for measurement and structural invariance. The aim of the present paper is to provide a tutorial in MG-CFA using the freely available R-packages lavaan, semTools, and semPlot. The combination of these packages enable a highly efficient analysis of the measurement models both for normally distributed as well as ordinal data. Data from two freely available datasets – the first with continuous the second with ordered indicators - will be used to provide a walk-through the individual steps.

Many researchers in psychology and social science are faced with the problem to compare latent constructs (i.e. mathematic ability, extraversion) that are not directly observable between different groups (languages, ethnic-groups), or points in time. Usually these latent constructs are measured by questionnaires, comprised of different scales that reflect different underlying latent variables. Typically differences between groups with regard to these underlying constructs are tested via scale means. Any comparison of means presuppose that the measures function similar in these different groups, i.e. that the response to individual items can be explained by the same latent factors (Byrne, Shavelson, & Muthén, 1989; Cheung & Rensvold, 1999; Vandenberg & Lance, 2000). Multiple-Group Confirmatory Factor Analysis (MG-CFA) has become the de-facto standard to investigate the degree to which measures are invariant across groups (Chen

important precursor for interpreting patient reported outcomes (Gregorich, 2006).

Although there are several introductions to MG-CFA to test for invariance that are based on commercial programs, e.g. AMOS (Byrne, 2004), several recent additions to the open source software R (R Core Team, 2012) enable researchers to perform such analysis with unprecedented efficiency. In this paper we will describe how the three packages lavaan (Rosseel, 2012), semPlot (Epskamp, 2013) and semTools can be combined to conduct MG-CFA analysis. Before providing a walk-through the analysis a short conceptual introduction is given.

A conceptual introduction to measurement invariance

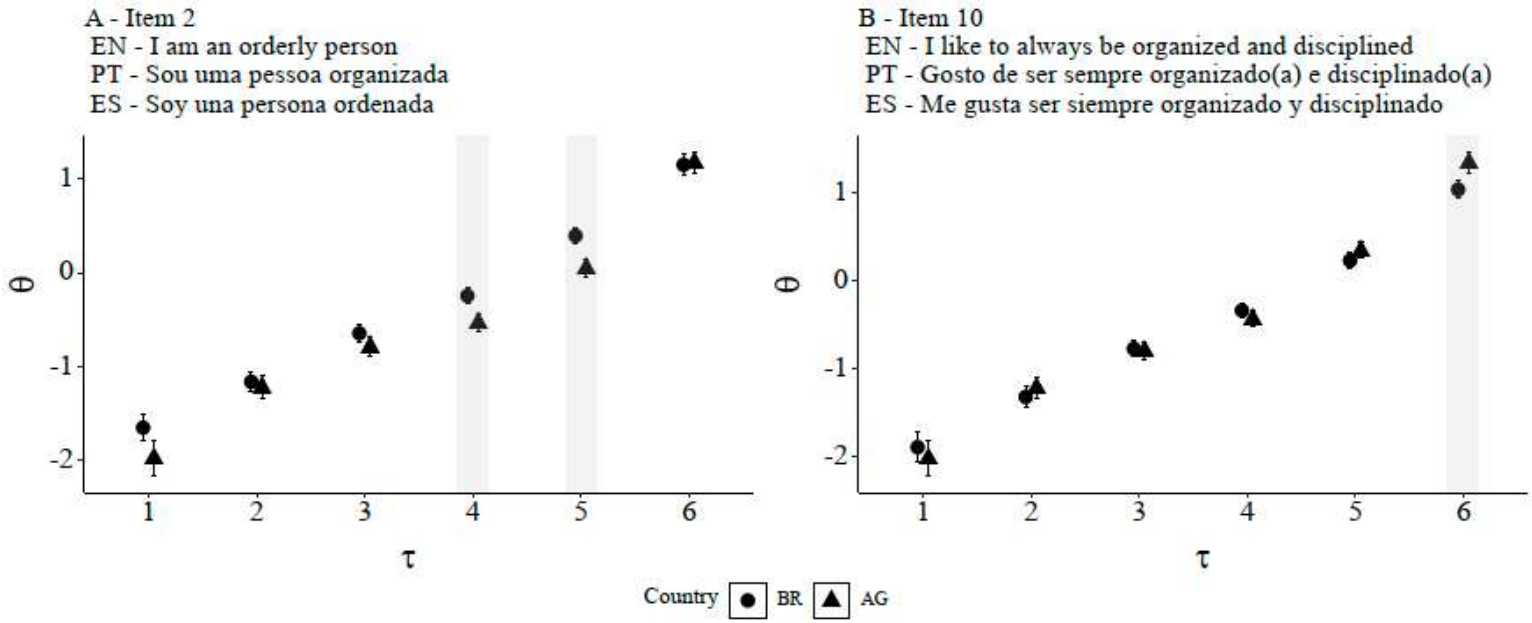
A scale is said to have measurement invariance

Invariância parcial

- Raramente temos invariância total dos instrumentos e isso não é um problema
  - Quer dizer, isso era o que gostaríamos, mas não é o fim do mundo ter um instrumento que não é 100% invariante
- Encontramos os parâmetros não invariantes por meio dos índices de modificação e *Lagrange Multiplier test*



Visualizando a não invariância invariância



O que veremos na prática?

- Invariância clássica de passos hierárquicos (o que eu acabei de mostrar)
- O que não veremos:
  - MIMIC
  - DIF a partir de teoria de resposta ao item
  - Técnicas bayesianas
  - Alinhamento (alignment)
  - Técnicas mistas (mixture models)
  - Técnica de Wu & Estabrook para itens ordinais

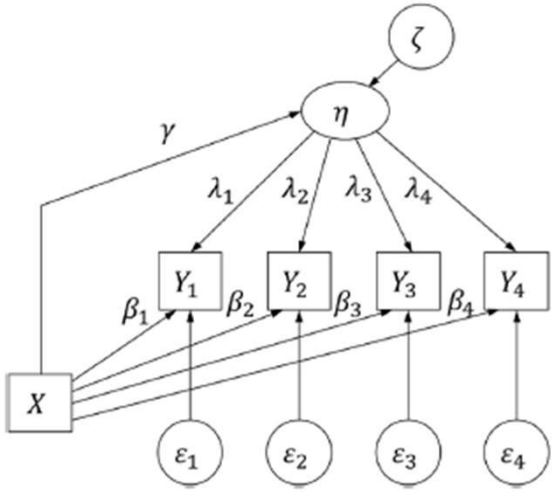


Fig. 2. The MIMIC DIF model with binary grouping variable.

Leituras sugeridas

Contents lists available at ScienceDirect

Social Science Research

journal homepage: www.elsevier.com/locate/ssresearch

Measurement invariance in the social sciences: Historical development, methodological challenges, state of the art, and future perspectives

Heinz Leitgöb [a,b,*], Daniel Seddig [c,d], Tihomir Asparouhov [e], Dorothée Behr [f], Eldad Davidov [c,g], Kim De Roover [h,i], Suzanne Jak [j], Katharina Meitinger [k], Natalja Menold [l], Bengt Muthén [e,m], Maksim Rudnev [n], Peter Schmidt [o,p], Rens van de Schoot [k]

[a] University of Leipzig, Germany
[b] University of Frankfurt, Germany
[c] University of Cologne, Germany
[d] University of Münster, Germany
[e] Mplus, USA
[f] GESIS – Leibniz Institute for the Social Sciences, Germany
[g] University of Zurich and URPP Social Networks, Switzerland
[h] Tilburg University, the Netherlands
[i] KU Leuven, Belgium
[j] University of Amsterdam, the Netherlands
[k] Utrecht University, the Netherlands
[l] Technische Universitaet Dresden, Germany
[m] University of California, USA
[n] University of Waterloo, Canada
[o] University of Giessen, Germany
[p] University of Mains, Germany

ARTICLE INFO

Keywords:
Comparative research
Measurement invariance
Item bias
Noninvariance detection
Multiple group confirmatory factor analysis
Scale construction

ABSTRACT

This review summarizes the current state of the art of statistical and (survey) methodological research on measurement (non)invariance, which is considered a core challenge for the comparative social sciences. After outlining the historical roots, conceptual details, and standard procedures for measurement invariance testing, the paper focuses in particular on the statistical developments that have been achieved in the last 10 years. These include Bayesian approximate measurement invariance, the alignment method, measurement invariance testing within the multilevel modeling framework, mixture multigroup factor analysis, the measurement invariance explorer, and the response shift-true change decomposition approach. Furthermore, the contribution of survey methodological research to the construction of invariant measurement instruments is explicitly addressed and highlighted, including the issues of design decisions, pretesting, scale adoption, and translation. The paper ends with an outlook on future research perspectives.

Contents lists available at ScienceDirect

Developmental Review

journal homepage: www.elsevier.com/locate/dr

Measurement invariance conventions and reporting: The state of the art and future directions for psychological research

Diane L. Putnick *, Marc H. Bornstein

Eunice Kennedy Shriver National Institute of Child Health and Human Development, Bethesda, MD, USA

ARTICLE INFO

Article history:
Received 4 November 2015
Revised 21 June 2016
Available online

Keywords:
Comparative psychology
Measurement invariance
Structural equation modeling
Confirmatory factor analysis
Multiple-group analysis

ABSTRACT

Measurement invariance assesses the psychometric equivalence of a construct across groups or across time. Measurement noninvariance suggests that a construct has a different structure or meaning to different groups or on different measurement occasions in the same group, and so the construct cannot be meaningfully tested or construed across groups or across time. Hence, prior to testing mean differences across groups or measurement occasions (e.g., boys and girls, pretest and posttest), or differential relations of the construct across groups, it is essential to assess the invariance of the construct. Conventions and reporting on measurement invariance are still in flux, and researchers are often left with limited understanding and inconsistent advice. Measurement invariance is tested and established in different steps. This report surveys the state of measurement invariance testing and reporting, and details the results of a literature review of studies that tested invariance. Most tests of measurement invariance include configural, metric, and scalar steps; a residual invariance step is reported for fewer tests. Alternative fit indices (AFIs) are reported as model fit criteria for the vast majority of tests; χ^2 is reported as the single index in a minority of invariance tests. Reporting AFIs is associated with higher levels of achieved invariance. Partial invariance is reported for about